Practice question on patterns

13 December 2025 12:26

If you decide to tackle these problems in Java, remember the fundamental steps:

1. Outer Loop: Controls the rows (usually from 1 to N).
2. Inner Loop(s): Control what gets printed in the current row (spaces, stars, numbers, or letters).
3. Print/Println: Use System.out.print() to keep the elements on the same line, and System.out.println() to move to the next line after a row is complete.

💻 7 Pattern Programming Questions (Nested Loops - Java)

1. Right Triangle Star Pattern (Basic Symbol)

Objective: Print a right-angled triangle using the asterisk (*) character.
Input: A number representing the number of rows (e.g., $N=5$).
Required Output:

```
*
* *
* * *
* * * *
* * * * *
```

Hint: Use two nested loops. The outer loop controls the rows (from 1 to N), and the inner loop controls the columns (from 1 up to the current row number).

2. Increasing Number Triangle Pattern

Objective: Print a triangle where each row contains the row number repeated.
Input: A number representing the number of rows (e.g., $N=4$).
Required Output:

```
1
2 2
3 3 3
4 4 4 4
```

Hint: The variable representing the **outer loop counter** (row number) is the number you need to print in the inner loop.

3. Palindromic Number Pyramid

Objective: Print a pyramid where numbers increase to the center and then decrease, forming a palindrome on each line.
Input: A number representing the number of rows (e.g., $N=5$).
Required Output:

```
    1
   121
  12321
 1234321
123454321
```

Hint: This requires three logical parts inside the outer loop:
1. Printing **spaces**.
2. Printing **increasing** numbers (from 1 up to the row number).
3. Printing **decreasing** numbers (from the row number - 1 down to 1).

4. Character Square Pattern (With Constant Character)

Objective: Print a square of characters, where a specific character is used for the

border and a different character for the inside.
Input: A number representing the side length (e.g., N=5).
Required Output (e.g., using 'A' for border and 'B' for inner):

```
A A A A A
A B B B A
A B B B A
A B B B A
A A A A A
```

Hint: Use an if condition inside the inner column loop to check if the current position is the **first row**, **last row**, **first column**, or **last column**.

5. Alphabetical Right Triangle Pattern (Increasing Letter)
Objective: Print a right-angled triangle where each row starts with 'A' and the letters increment across the row.
Input: A number representing the number of rows (e.g., N=5).
Required Output:

```
A
A B
A B C
A B C D
A B C D E
```

Hint: Initialize a character variable, say char c = 'A', and print its value. Increment this character variable (c++) in the inner loop. Reset it to 'A' at the start of each row (before the inner loop).

6. Hollow Diamond Pattern
Objective: Print a diamond shape that is hollow inside (only the borders are printed).
Input: A number representing the maximum number of stars in the middle row (must be odd, e.g., N=7).
Required Output (for N=7):

```
   *
  * *
 *   *
*     *
 *   *
  * *
   *
```

Hint: Divide the pattern into two halves: the **upper half** (increasing rows) and the **lower half** (decreasing rows). In each half, you'll need to print spaces, then a star, then more spaces (for the hollow part), and finally another star.

7. Zero-One Triangle (Binary Pattern)
Objective: Print a triangle where the numbers are 0 and 1, alternating in a specific pattern (e.g., the sum of the row and column index determines the value).
Input: A number representing the number of rows (e.g., $N=5$).
Required Output:

```
1
0 1
1 0 1
0 1 0 1
1 0 1 0 1
```

Hint: Use the outer loop counter(i) and the inner loop counter(j). The number to print is 1 if the sum(i+j) is **even**, and '0' if the sum(i+j) is **odd**. (Adjust indices based on

whether you start counting from 0 or 1).